SHRI VAISHNAV VIDYAPEETH VISHWAVIDYALAYA, INDORE

Shri Vaishnav Institute of Information Technology (SVIIT) • Department of Computer Science & Engineering

CASE MANAGEMENT HEARING (CMH) e-PORTAL

STATUTORY PRE-TRIAL JUDICIAL WORKSPACE UNDER ORDER XV-A, CODE OF CIVIL PROCEDURE (CPC)

Academic Course: Major Project (BTCS706N / Semester VII-VIII)
Project Guide: Prof. Shweta Pandey
Target Bench: High Court of Madhya Pradesh (Indore, Jabalpur, Gwalior)

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1. EXECUTIVE SUMMARY & RESEARCH FOUNDATION

> *"Our project is an active judicial workspace that transforms court management from a passive electronic noticeboard into an active, collaborative digital courtroom. It digitizes pre-trial proceedings under Order XV-A of the CPC to eliminate advocate scheduling clashes, automate admissions of evidence, and slash pre-trial delays by up to 50%."*

The 53-Million Backlog Crisis: Over 53 million cases remain pending across Indian courts. A landmark 2024 empirical study published by **Cambridge University Press**, conducted by researchers from **IISc Bangalore and NLSIU across 1,775 district courts** (Aithala et al.), proved an eye-opening reality: *merely appointing more judges does not reduce court backlogs*. Over 50% of civil suits stagnate for a median of **394 days in preliminary pre-trial and appearance stages** alone.

Passive Noticeboard vs. Active Workspace Innovation:
Existing systems like `eCourts Services` and `mphc.gov.in` are **passive noticeboards**—they only publish past static PDF cause lists and scanned orders. They lack any online workspace for lawyers to negotiate undisputed facts, cross-check advocate availability, or conduct structured pre-trial case management conferences. Our portal bridges this critical gap.

2. STATUTORY & PROCEDURAL FRAMEWORK

STATUTORY PROVISION	PROCEDURAL REQUIREMENT IN LAW	IMPLEMENTATION IN CMH E-PORTAL
Order XV-A Rule 1 CPC	Mandatory Case Management Hearing held within 4 weeks of written statement filing.	Automated pre-trial hearing scheduling engine with strict compliance date monitors.
Order XI Rule 4 CPC	Filing statement of Admissions & Denials of Documents.	Interactive digital checklist where lawyers accept/deny contracts, invoices, and emails online.
Order XV-A Rule 6 CPC	Presiding Judge must issue Case Management Order (CMO) fixing fixed trial timetable.	1-Click automated bilingual judicial order generator with digital signature authentication.
Order V Rule 9 CPC	Service of summons by electronic transmission and modern delivery methods.	Multi-channel automated notice dispatcher (In-App, SMS, WhatsApp, and Email).

STATUTORY PROVISION	PROCEDURAL REQUIREMENT IN LAW	IMPLEMENTATION IN CMH E-PORTAL
Section 89 CPC	Court-mandated referral to Alternative Dispute Resolution (ADR).	Automated ADR referral matrix to Mediation, Conciliation, Lok Adalat, or Arbitration.
Sec 65B Indian Evidence Act	Admissibility of electronic records and tamper-proof certificate verification.	Every uploaded exhibit receives an immutable **SHA-256 cryptographic hash** in the Judicial Vault.
MP Court Fees Act	Mandatory ad-valorem court fee payment prior to judicial registration.	Cyber Treasury e-Challan module with dynamic UPI QR code generation and verification.

3. THE 7 CORE ARCHITECTURAL PILLARS

1. Universal Case Dossier & CNR Tracking

CORE REPOSITORY

Provides a single-window search across all commercial cases in Madhya Pradesh using Case Number Record (CNR) format (e.g. `MPGW010001112026`). Features an interactive 6-stage case progress tracker, complete pleadings audit, and ad-valorem claim valuation summaries.

2. Digital Admissions & Denials Matrix

ORDER XI RULE 4 CPC

Replaces traditional 100-page paper affidavits. Counsel for Plaintiff and Defendant check off digital exhibits with specific legal stances: *"Admitted as to execution"*, *"Admitted as to contents"*, or *"Denied as to valuation"*. Admitted facts are instantly excluded from trial scope, and contested items automatically synthesize into **Framed Issues for Trial**.

3. Conflict-Free Smart Scheduler (Clash Detector)

ZERO DOUBLE-BOOKING

Prevents cross-courtroom advocate double-booking. When a clerk lists a CMH hearing, the backend algorithm analyzes the advocate's Bar Council ID across all courtroom halls and judicial benches. If a time clash is detected, the clerk is warned immediately and conflict-free alternative slots are suggested.

4. In-Browser WebRTC Virtual Courtroom

ORDER XV-A PRACTICE DIRECTIONS

Integrated encrypted video conferencing powered by WebRTC & Jitsi Meet. Enforces strict 15-minute scheduled lock/unlock admittance rules. Includes a split-screen judicial view displaying high-definition participant video feeds on the left while simultaneously reviewing SHA-256 tamper-proof evidence on the right.

5. 1-Click Automated Case Management Order (CMO) Generator

ORDER XV-A RULE 6 CPC

Synthesizes hearing outcomes, framed issues, admitted evidence, discovery deadlines, and witness limits into an official, digitally signed Case Management Order PDF in under 60 seconds, eliminating multi-day stenographer typing delays.

6. Cyber Treasury e-Challan & UPI QR Payment

COURT FEES ACT

Automated court fee calculation based on suit valuation. Issues official government Major Head (`0030-02-103-01`) challans with dynamic UPI QR code generator, real-time treasury verification, and printable legal receipts.

7. Statutory Multi-Channel Notification Service

ORDER V CPC

Automated dispatch engine delivering digital summons, hearing reminders, and CMO notices via In-App notifications, SMS, and WhatsApp alerts with verifiable delivery logs.

4. SOFTWARE ENGINEERING & DESIGN PATTERNS (GANG OF FOUR)

DESIGN PATTERN	SOFTWARE IMPLEMENTATION LOCATION	REAL-WORLD ARCHITECTURAL PURPOSE
Strategy Pattern	`backend/app/patterns/strategy.py`	Interchangeable slotting algorithms: *Strict Strategy* (zero tolerance), *Buffered Strategy* (15-min transit buffer), and *Virtual Hybrid Strategy* (relaxed hall conflict for WebRTC).
Singleton Pattern	`backend/app/patterns/singleton.py`	Thread-safe, global clash detection instance and database session managers ensuring single-source-of-truth across concurrent requests.
Factory Method	`backend/app/patterns/factory.py`	Decoupled creation of case entities (Commercial Dispute, Summary Suit, Arbitration) and statutory documents (Summons, CMO, e-Challans).

DESIGN PATTERN	SOFTWARE IMPLEMENTATION LOCATION	REAL-WORLD ARCHITECTURAL PURPOSE
Observer Pattern	`backend/app/routers/notifications.py`	Publish-subscribe event model dispatching real-time notifications to In-App, SMS, and WhatsApp subscribers upon case state transitions.

5. TECHNOLOGY STACK & TECHNICAL SPECIFICATIONS

LAYER	TECHNOLOGIES USED	ENGINEERING JUSTIFICATION
Frontend	React 19, Vite, Tailwind CSS, Lucide Icons	Sub-second HMR, high responsiveness, dark/light theme for judicial accessibility.
Backend API	Python 3.11 / 3.13, FastAPI, Pydantic v2	Asynchronous ASGI execution, native `async/await`, built-in Swagger OpenAPI docs at `/docs`.
Database	SQLite (Local) / MySQL 8.0 (Production), SQLAlchemy ORM	ACID compliance for legal dockets, relational integrity, native JSON support for checklists.
Video Conferencing	HTML5 WebRTC, Jitsi Meet External API	Zero-install in-browser video hearings, screen sharing, end-to-end media encryption.
Security	JWT Bearer Tokens, PBKDF2/Bcrypt, SHA-256	Role-Based Access Control (RBAC), tamper-proof cryptographic evidence vault.

6. ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

USER ROLE	AUTHORIZED CAPABILITIES & WORKFLOWS
Judge (न्यायाधीश)	Presides over CMH conferences, convenes/concludes virtual hearings, reviews Admissions Checklist, rules on disputed facts, digitally signs Case Management Orders (Order XV-A Rule 6).
Advocate (अधिवक्ता)	Files petitions and written statements, completes Admissions & Denials checklists, joins virtual hearings, pays statutory e-challans via UPI QR, receives electronic summons.
Registry Clerk (न्यायालय लिपिक)	Scrutinizes new filings, allocates clash-free court slots, manages daily cause lists, verifies Cyber Treasury e-challans, issues electronic summons under Order V.
Litigant / Citizen (नागरिक)	Real-time suit tracking by CNR number, reviews public cause list, downloads verified court orders, views fee receipts and transparent admissions matrices.

7. PANEL VIVA QUESTIONS & WINNING MODEL ANSWERS

Q1: What exactly is a Case Management Hearing (CMH)? How does it differ from a trial hearing?

Answer: A regular trial hearing is where witnesses are cross-examined and final arguments take place. A Case Management Hearing (CMH) is an early pre-trial conference conducted under Order XV-A CPC. Its purpose is to filter out agreed facts, resolve document disputes, set strict discovery deadlines, and explore mediation *before* trial starts. This ensures zero court time is wasted on routine paperwork when the trial begins.

Q2: Why can't existing portals like eCourts Services or the MP High Court website (mphc.gov.in) do this?

Answer: Current court portals are passive electronic noticeboards. They only display past case statuses, uploaded orders, and static cause list PDFs. They have no interactive pre-trial workspace: they cannot allow lawyers to settle admissions online, they cannot detect advocate schedule clashes across multiple courtrooms, and they cannot automate case management orders.

Q3: What empirical research supports this project? What did the 2024 Cambridge study prove?

Answer: Our project is backed by a landmark 2024 empirical study published by Cambridge University Press (Aithala et al. from IISc Bangalore & NLSIU) analyzing 1,775 district courts. They proved that appointing more judges does not solve backlogs because over 50% of civil suits stagnate for a median of 394 days in preliminary pre-trial appearance stages. Active pre-trial case management is the single biggest factor in reducing delay.

Q4: How does the Smart Scheduling algorithm prevent advocate double-booking?

Answer: Every advocate has a unique State Bar Council ID linked to their appearance. When a clerk schedules a hearing, our backend algorithm checks the advocate's global schedule across all court halls and benches. If an overlap or transit conflict is detected, the system immediately flags the clash and suggests conflict-free alternative time slots.

Q5: Why did you choose Python FastAPI over Django, Node.js, or PHP?

Answer: FastAPI is built on ASGI (Starlette) and supports native Python `async/await`, delivering performance comparable to Go and Node.js while leveraging Python's rich data ecosystem. It also provides automatic Pydantic v2 data validation—essential for validating complex legal inputs—and generates interactive Swagger OpenAPI documentation at `/docs`.

Q6: How does document security work under Section 65B of the Indian Evidence Act?

Answer: Whenever an evidence document or contract is uploaded, our backend generates an immutable SHA-256 cryptographic hash stored in the Judicial Vault. During virtual hearings, both parties and the judge can verify that the hash matches the original exhibit, proving mathematically that the evidence has not been tampered with.